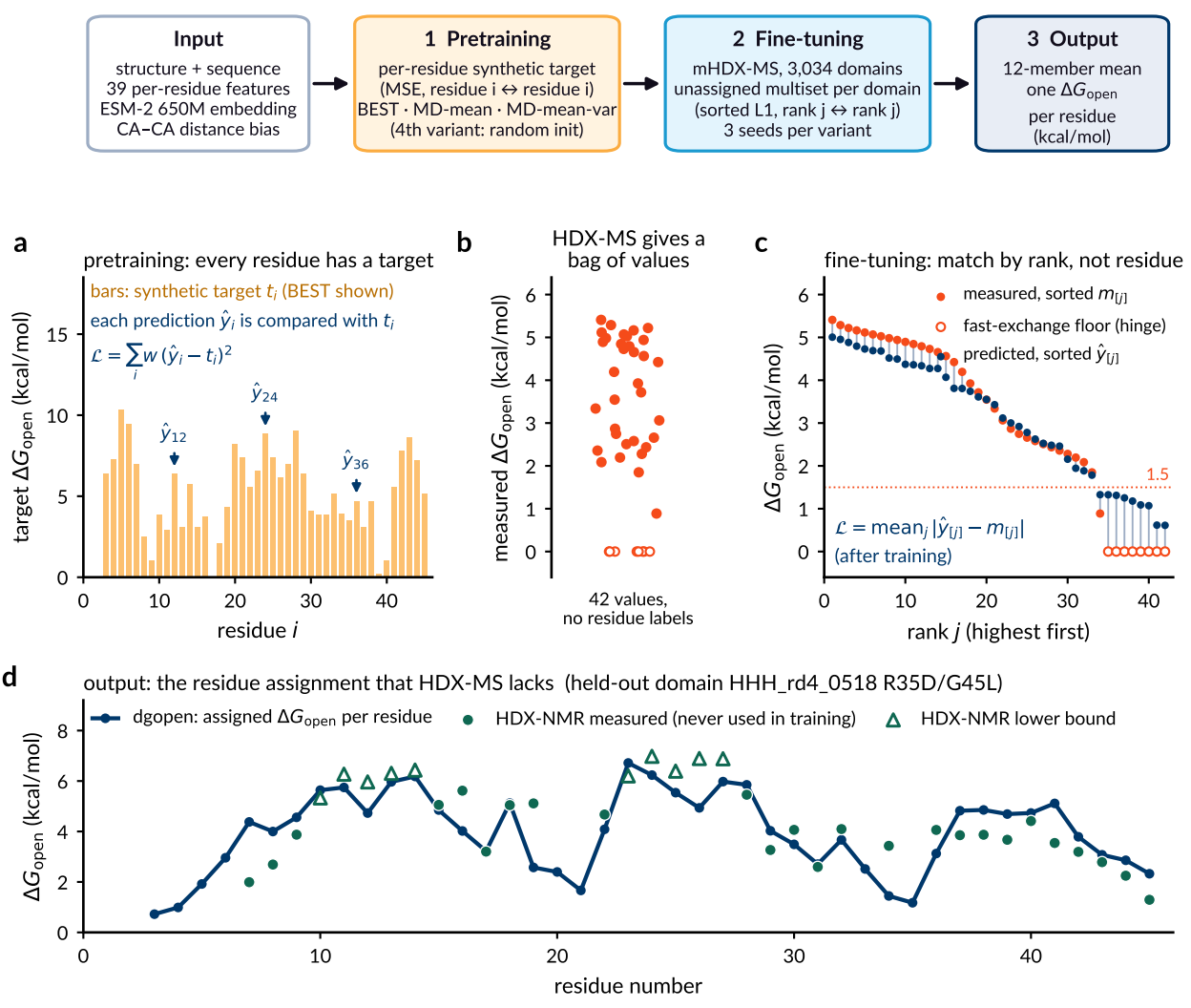

Input
structure + sequence
39 per-residue features
ESM-2 650M embedding
CA–CA distance bias
1 Pretraining
per-residue synthetic target
(MSE, residue i ⇔ residue i)
BEST · MD-mean · MD-mean-var
(4th variant: random init)
2 Fine-tuning
mHDX-MS, 3,034 domains
unassigned multiset per domain
(sorted L1, rank j ⇔ rank j)
3 seeds per variant
3 Output
12-member mean
one ΔG_{open}
per residue
(kcal/mol)
a
pretraining: every residue has a target
bars: synthetic target t_i (BEST shown)
each prediction $\hat{y}_i$ is compared with t_i
$\mathcal{L} = \sum_i w\,(\hat{y}_i - t_i)^2$
$\hat{y}_{12}$
$\hat{y}_{24}$
$\hat{y}_{36}$
target ΔG_{open} (kcal/mol)
residue i
b
HDX-MS gives a bag of values
measured ΔG_{open} (kcal/mol)
42 values,
no residue labels
c
fine-tuning: match by rank, not residue
measured, sorted $m_{[j]}$
fast-exchange floor (hinge)
predicted, sorted $\hat{y}_{[j]}$
1.5
$\mathcal{L} = \text{mean}_j\,|\hat{y}_{[j]} - m_{[j]}|$
(after training)
ΔG_{open} (kcal/mol)
rank j (highest first)
d
output: the residue assignment that HDX-MS lacks (held-out domain HHH_rd4_0518 R35D/G45L)
dgopen: assigned ΔG_{open} per residue
HDX-NMR measured (never used in training)
HDX-NMR lower bound
ΔG_{open} (kcal/mol)
residue number